AGES 6–12 | COMPLETE BEGINNER FRIENDLY | FOR FIRST-TIME CODERS

CURRICULUM OVERVIEW

Little Coders

Scratch, AI & Python — One Playful Step at a Time

A gentle, hands-on journey: from colourful Scratch blocks to games, AI-powered projects and block-based Python — building real, shareable projects the whole way. Five independent courses; families enrol one course at a time.



40 LESSONS · 5 COURSES
Scratch, AI and Python — each lesson ends in a project

6 ASSESSMENTS
5 course assessments + 1 final assessment

5	40	46	6–12
Courses	Lessons	Classes in total	Age range

WHO IT IS FOR
For first-time coders — complete beginners welcome

BEFORE YOU START
No coding experience needed

LEARNING PROGRESSION

01 Scratch Adventures
Coding fundamentals, animations, stories and sprites

02 Scratch Game Studio
Advanced Scratch, game mechanics and interactive projects

03 AI for Kids: Smart Projects
Image and sound recognition with AI-powered Scratch projects

04 Python Explorers
Introduction to Python using visual blocks

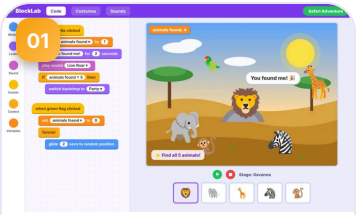
05 Python Creative Lab
Intermediate block-based Python and independent projects

Learn by making
Every lesson has a mission, a demo and a project — children build something they can open, share and show.

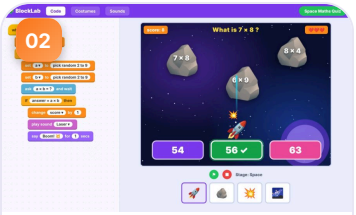
One course at a time
The five courses are independent: start with Scratch Adventures, or pick up where your child left off.

Assessments that feel like a game
Think in blocks, a quick check, then build it: friendly 40-minute assessments, with no negative marking.

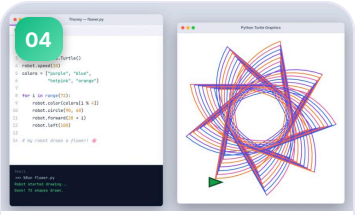
HANDS-ON PROJECTS STUDENTS BUILD



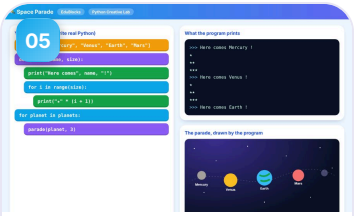
Safari Adventure
Scratch-style blocks



Space Maths Game
Scratch-style blocks



Drawing Robot
Python · Turtle



Space Parade
EduBlocks · Python

COURSE 01 · LESSONS 1–8

01

Scratch Adventures

Learn how code works by launching rockets, dancing aliens and telling your first animated story in MIT Scratch.

MIT Scratch

8 REAL PROJECTS

1 COURSE ASSESSMENT

OVERVIEW

Coding fundamentals, animations, stories and sprites. Build scripts that run in the right order; Move, animate and talk with sprites — and more.

PORTFOLIO PROJECTS

- Rocket Launch
- Dancing Alien
- Talking Space Pet
- Control Your Rocket
- Space DJ
- Dancing Starfield
- Alien Animation Studio
- Lost Space Puppy

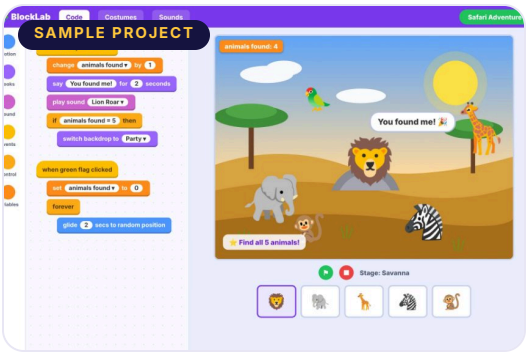
WHY IT MATTERS

Scratch turns “how does code work?” into play. Children learn to give instructions in order, make characters move and talk, and finish a story — the foundation of every later project.

LESSON BREAKDOWN

1. Welcome to Scratch!
2. Let's Make Things Move!
3. Give Your Characters a Personality
4. Making Things Interactive
5. Sounds, Music & Effects
6. The Magic of Repetition
7. Bringing Characters to Life
8. My First Animated Story

8 LESSONS — TEST UNLOCKS



HANDS-ON PROJECT

Safari Adventure

Scratch-style blocks

A story project where sprites move, talk and react — the animation and event ideas of this course in action.

COURSE ASSESSMENT

Scratch Adventures — Course Assessment

A friendly check on what you built in Scratch Adventures: think in blocks, a quick check, then build a project. Taken after lesson 8.

10 multiple-choice · 5 true / false · 25-mark project · 40 minutes

COURSE 02 · LESSONS 9–16

02 Scratch Game Studio

Turn Scratch skills into real games: controls, collisions, scores, lives, levels and clones.

MIT Scratch

8 REAL PROJECTS

1 COURSE ASSESSMENT

OVERVIEW

Advanced Scratch, game mechanics and interactive projects. Control a sprite with the keyboard inside a maze; Detect collisions and keep score — and more.

PORTFOLIO PROJECTS

- Asteroid Maze
- Catch the Falling Stars
- Star Collector
- Keycard Escape
- Mystery Prize Machine
- Meteor Dodge
- Space Mission: Three Levels
- Meteor Blaster

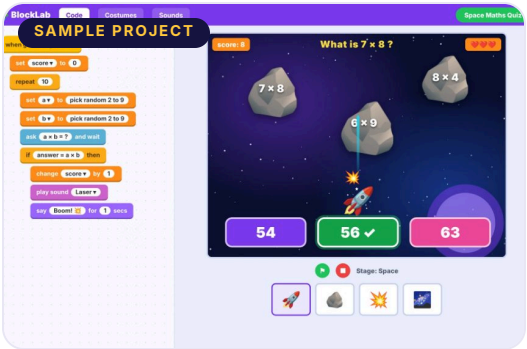
WHY IT MATTERS

Games are the best reason to practise logic. Controls, scores, lives and levels teach children to plan, test and fix — and they end with a game friends can play.

LESSON BREAKDOWN

1. Game Controls & Movement
2. Making Sprites Interact
3. Keeping Score!
4. Making Decisions with Code
5. Randomness & Game Challenges
6. Timers, Lives & Survival
7. Multiple Levels & Game Scenes
8. Clones & Arcade Games

8 LESSONS — TEST UNLOCKS



HANDS-ON PROJECT

Space Maths Game

Scratch-style blocks

A game with controls, a score and a challenge, built from sprites, variables and decisions.

COURSE ASSESSMENT

Scratch Game Studio — Course Assessment

A friendly check on what you built in Scratch Game Studio: think in blocks, a quick check, then build a project. Taken after lesson 8.

10 multiple-choice · 5 true / false · 25-mark project · 40 minutes

COURSE 03 · LESSONS 17–24

03 AI for Kids: Smart Projects

Train your own AI with a webcam and a microphone, then bring it to life in Stretch3 with the TM2Scratch extension.

Generation AI Teachable Machine | Stretch3 | TM2Scratch extension

8 REAL PROJECTS

1 COURSE ASSESSMENT

OVERVIEW

Image and sound recognition with AI-powered Scratch projects. Train an image model with everyday objects; Train a model that recognises spoken words — and more.

PORTFOLIO PROJECTS

- Alien Pet Café
- Pico's Rescue Mission
- Recycling Station
- Superhero Power-Up Lab
- Space Explorer Packing Mission
- Voice-Controlled Rocket
- The Wizard's Magic Spellbook
- The AI Quiz Show

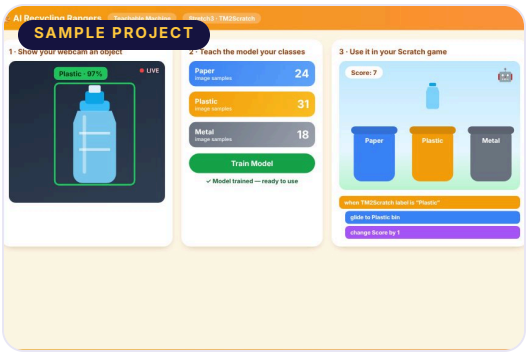
WHY IT MATTERS

Children use AI every day without seeing how it learns. Here they train their own model with a webcam and a microphone, so AI stops being magic and becomes something they understand.

LESSON BREAKDOWN

1. Giga's AI Snack Buddy
2. Pico's AI Space Rescue Scanner
3. AI Recycling Rangers
4. AI Superhero Power-Up Lab
5. AI Space Explorer Packing Mission
6. AI Voice-Controlled Rocket Mission
7. The Wizard's Magic Spellbook
8. The AI Quiz Show Challenge

8 LESSONS — TEST UNLOCKS



HANDS-ON PROJECT

AI Recycling Rangers

Teachable Machine · Stretch3

A sorting game powered by a model the student trained themselves on everyday objects with a webcam.

COURSE ASSESSMENT

AI for Kids: Smart Projects — Course Assessment

A friendly check on what you built in AI for Kids: Smart Projects: think in blocks, a quick check, then build a project. Taken after lesson 8.

10 multiple-choice | 5 true / false | 25-mark project | 40 minutes

COURSE 04 · LESSONS 25–32

04 Python Explorers

Take your first steps in Python with EduBlocks: print, draw with Turtle, repeat, calculate, ask questions and decide.

EduBlocks | Python

8 REAL PROJECTS

1 COURSE ASSESSMENT

OVERVIEW

Introduction to Python using visual blocks. Read and build Python programs with blocks; Draw shapes and patterns with loops — and more.

PORTFOLIO PROJECTS

- Mission Control Console
- Rocket Badge
- Rainbow Galaxy
- Moon Base Supply Calculator
- Space Passport Generator
- Oxygen Safety Checker
- Alien Treasure Machine
- Space Rescue Quiz

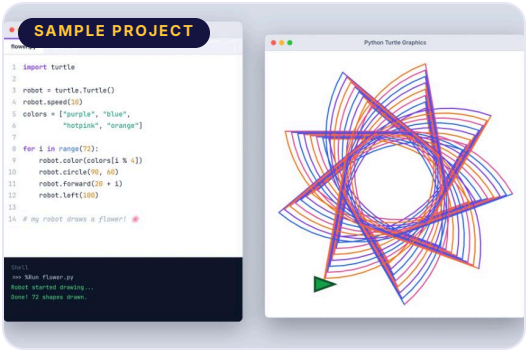
WHY IT MATTERS

Block-based Python gives children real Python ideas — loops, variables, decisions — without the typing hurdle, so the move to typed code later feels natural.

LESSON BREAKDOWN

1. Welcome to Python!
2. Drawing with Python
3. Repeat Magic in Python
4. Variables & Calculations
5. Let the User Join In!
6. Making Decisions in Python
7. Randomness & Smart Decisions
8. My First Python Game

8 LESSONS — TEST UNLOCKS



HANDS-ON PROJECT

Drawing Robot

Python · Turtle

Python drawing with loops: code that makes the turtle trace a pattern, shown next to the shape it draws.

COURSE ASSESSMENT

Python Explorers — Course Assessment

A friendly check on what you built in Python Explorers: think in blocks, a quick check, then build a project. Taken after lesson 8.

10 multiple-choice | 5 true / false | 25-mark project | 40 minutes

COURSE 05 · LESSONS 33–40

05 Python Creative Lab

Level up with lists, functions, while loops, nested loops and debugging — then build your own Python space adventure.

EduBlocks · Python

8 REAL PROJECTS

1 COURSE ASSESSMENT

OVERVIEW

Intermediate block-based Python and independent projects. Store and use collections of values; Write your own functions with parameters — and more.

PORTFOLIO PROJECTS

- Planet Picker
- Solar System Parade
- Star Stamp Studio
- Custom Planet Maker
- Space Password Challenge
- Galaxy Pattern Generator
- Repair Mission Control
- Space Explorer Adventure

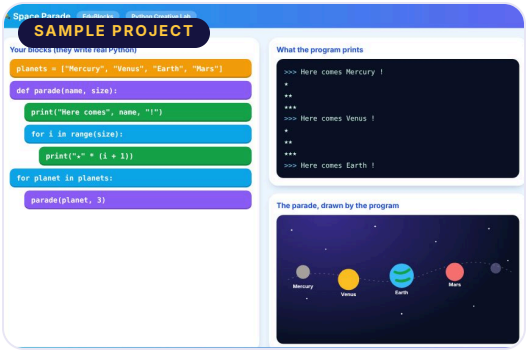
WHY IT MATTERS

Lists, functions and bug-hunting are the habits of real programmers. This course lets children use them on their own space adventure, independently.

LESSON BREAKDOWN

1. Working with Lists
2. Loops & Collections
3. Creating Your Own Functions
4. Functions with Superpowers
5. Keep Going Until It Works!
6. Loops Inside Loops
7. Finding & Fixing Bugs
8. My Python Capstone

8 LESSONS — TEST UNLOCKS



HANDS-ON PROJECT

Space Parade

EduBlocks · Python

A block-based Python program with lists, functions and loops that runs a whole space parade.

COURSE ASSESSMENT

Python Creative Lab — Course Assessment

A friendly check on what you built in Python Creative Lab: think in blocks, a quick check, then build a project. Taken after lesson 8.

10 multiple-choice · 5 true / false · 25-mark project · 40 minutes

HOW STUDENTS ARE TESTED · 6 TESTS

Level-Up Checkpoints

Each of the five courses ends with an assessment, and a final assessment closes the program. They are 40 minutes and 40 marks, in a friendly layout: think in blocks, a quick check, then a build-it project. No negative marking, ever.



THE CHECKPOINT PATH — EVERY COURSE ENDS WITH A TEST, THEN ONE FINAL

Course 1 · 8 lessons → COURSE 1 → Course 2 · 8 lessons → COURSE 2 → Course 3 · 8 lessons → COURSE 3 → Course 4 · 8 lessons → COURSE 4 → Course 5 · 8 lessons → COURSE 5 → FINAL

COURSE ASSESSMENT · after every 8 lessons

Think, check and build

Every course ends with an assessment in the same friendly layout: think in blocks, a quick check, then build a project of your own.

10	Think in Blocks — multiple-choice questions, many with a block script
5	Quick Check — true / false
25	Build It! — one practical project in Stretch3 (courses 1–3) or EduBlocks (4–5)

FINAL ASSESSMENT · the finish line

One assessment on the whole program

The finish line: the same three parts, drawing on all five courses — with a Scratch & AI project or a Python project.

10	Think in Blocks — multiple-choice questions, many with a block script
5	Quick Check — true / false
25	Build It! — one practical project in Stretch3 (courses 1–3) or EduBlocks (4–5)

A TASTE OF A TEST

SAMPLE · MULTIPLE-CHOICE

Which block makes a sprite repeat the same steps again and again?

- A forever ✓ CORRECT
- B say Hello!
- C play sound Meow
- D go to x: 0 y: 0

SAMPLE · TRUE / FALSE

In Python, print() shows words on the screen.

True ✓ | False

Examples in the style of a real test — written for this document, not taken from one.

AFTER EVERY TEST

Marked and checked
The quiz parts are marked automatically. A teacher marks the Build It! project against a clear rubric.

Shared with parents
A report reaches parents once the teacher has completed the review.

Never a penalty for guessing
There is no negative marking, and the last five minutes are kept for review and hand-in.

5 course tests • 1 final test • Every test is part of the program

The Complete Journey

Across 5 courses, 40 lessons and 6 assessments, children move from their first Scratch sprite to AI projects and block-based Python — about six months at two classes a week, and a real project at the end of every lesson.

5 COURSES · 40 LESSONS

01 Scratch Adventures
Dancing Alien · Space DJ · Lost Space Puppy
8 lessons · + course assessment

02 Scratch Game Studio
Catch the Falling Stars · Mystery Prize Machine · Meteor Blaster
8 lessons · + course assessment

03 AI for Kids: Smart Projects
Pico's Rescue Mission · Space Explorer Packing Mission · The AI Quiz Show
8 lessons · + course assessment

04 Python Explorers
Rocket Badge · Space Passport Generator · Space Rescue Quiz
8 lessons · + course assessment

05 Python Creative Lab
Solar System Parade · Space Password Challenge · Space Explorer Adventure
8 lessons · + course assessment

HOW STUDENTS ARE TESTED · 6 TESTS

COURSE ASSESSMENT 1 — Scratch Adventures
COURSE ASSESSMENT 2 — Scratch Game Studio
COURSE ASSESSMENT 3 — AI for Kids: Smart Projects
COURSE ASSESSMENT 4 — Python Explorers
COURSE ASSESSMENT 5 — Python Creative Lab

FINAL TEST
One assessment on the whole program
The finish line: the same three parts, drawing on all five courses — with a Scratch & AI project or a Python project.

BY THE END, STUDENTS CAN

- Build animated stories and arcade-style games in Scratch.
- Train an AI model with a webcam and a microphone, and use it in a Scratch project.
- Read and build Python programs with blocks, then finish a space adventure of their own.

Ready for the first mission?
All lessons and projects are taught live by an expert Lumexa instructor.
Lumexa Little Coders · Ages 6–12